INDIA'S IMPORT TRADE WITH ASIAN COUNTRIES

A Data-Driven Analysis of Trade Trends and Patterns (2015–2025)

Python Data Analytics Project

1. Project Overview

This project analyzes India's import trade with Asian countries from 2015 to 2025 using Python. The analysis focuses on major importing countries, high-value commodities, yearly import trends, Asian trade subregional patterns, measurement-unit patterns and the statistical distribution of import values.

The project follows an end-to-end data analytics workflow covering data import, exploration, data-quality assessment, cleaning, transformation, feature engineering, statistical analysis, visualization, interpretation and decision-support insights.

The dataset contains more than three million trade records, providing practical experience in handling large-scale structured data and extracting meaningful patterns.

2. Problem Statement

Large import-trade datasets contain millions of records across dates, countries, commodities, geographical classifications and measurement units. Individual records do not easily reveal major contributors, long-term trends or unusual patterns.

This project applies exploratory and statistical analysis to summarize the data, identify important trade partners and commodities, compare patterns over time and geography, and highlight observations requiring further validation.

3. Objectives

- Identify major importing countries based on import value and trade activity.
- Identify major commodities contributing to import activity.
- Analyze import trends from 2015 to 2025.
- Compare import activity across countries and Asian trade subregions.
- Analyze the distribution and variation of import values using statistical measures.
- Generate data-driven insights to support trade monitoring and decision-making.

4. Dataset & Tools

Source: India Data Portal

Scope: India's imports from Asian countries

Period: 2015–2025

Initial dataset: 3,188,530 records × 15 columns.

Final cleaned dataset: 3,188,475 records.

After feature engineering: 21 features.

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn and Google Colab.

The dataset contains identifiers, dates, country and geographical classifications, HS-code/commodity information, measurement units, quantity and import values in Indian Rupees and US Dollars.

5. Tools and Technologies

Python – overall analysis and processing.

Pandas – data loading, cleaning, grouping and aggregation.

NumPy – numerical operations and calculations.

Matplotlib – data visualization.

Seaborn – statistical and distribution visualization.

Google Colab – development and execution environment.

6. Data Understanding and Exploratory Data Analysis

The initial analysis examined dataset dimensions, columns, data types, missing values, duplicates and unique categorical values. Country, commodity, date, subregion and measurement-unit fields were reviewed to understand the structure of the trade data.

6.1 Dataset Structure

- 3,188,530 initial rows and 15 columns.
- Trade-level observations across countries, commodities, dates and units.
- 55 records with missing measurement units were removed.
- Final analytical dataset: 3,188,475 records.

6.2 Important Variables

- date – trade date used for time analysis.
- country_name – country classification.
- region_name – regional classification; analysed records belong to Asia.
- subregion_name – Asian subregion used for geographic comparison.
- hs_code – commodity classification code.
- commodity – commodity description.
- unit – measurement unit.
- value_qt – reported quantity.
- value_rs – import value in Indian Rupees; primary analytical variable.
- value_dl – import value in US Dollars.

6.3 Exploratory Observations

Import activity is not evenly distributed across countries or commodities. A relatively small number of countries and high-value commodities contribute a substantial share of total import value. The subregion field provides useful geographic variation for comparison.

7. Data Cleaning and Preprocessing

7.1 Missing Values

Missing-value checks were performed across the dataset. Fifty-five records had missing measurement units and were removed because the unit field is important for interpreting quantity-related information.

7.2 Duplicate Records

Duplicate checks were performed to identify repeated records. The final cleaned dataset contains 3,188,475 records.

7.3 Standardization
The country name “Türkiye” was standardized to “Turkey” so country-level grouping would remain consistent.

7.4 Date Transformation
The date field was transformed to derive Year and Month-related analytical features for time-based analysis.

7.5 Zero-Value Analysis
Zero-value checks were performed for quantity and import-value fields. Zero observations were analyzed rather than automatically removed because they may represent valid source records.

7.6 Selection of Analytical Variable
value_rs, representing import value in Indian Rupees, was selected as the primary analytical variable. value_dl is a redundant currency representation. value_qt contains incompatible units such as Kgs, Nos and Sqm; therefore, overall quantity aggregation across units was not treated as analytically meaningful.

8. Feature Engineering
- Derived Year from date for yearly trend analysis.
- Created month-related fields for time-based analysis.
- Standardized country names for consistent grouping.
- Created aggregated tables for yearly, country, commodity, unit and subregion comparisons.
- Applied log transformation for distribution visualization because import values are highly right-skewed.

9. Statistical Analysis
Descriptive statistics were calculated for value_rs, the primary import-value variable.

Measure	Result
Mean	₹699.02
Median	₹18.42
Mode	₹0
Variance	207,986,172.44
Standard Deviation	₹14,421.73
Skewness	90.41
Kurtosis	11,190.26

9.1 Interpretation
- Mean: The average import value per trade record is approximately ₹699.02.
- Median: The median is ₹18.42, showing that typical records have much lower values than the average.
- Mode: The mode is ₹0, indicating frequent zero-value observations.
- Variance: The large variance indicates substantial dispersion in import values.
- Standard Deviation: The standard deviation of approximately ₹14,421.73 shows wide variation among trade values.
- Skewness: The skewness of 90.41 indicates a very strong right-skewed distribution with a long upper tail.
- Kurtosis: The high kurtosis indicates extreme observations and heavy tails.

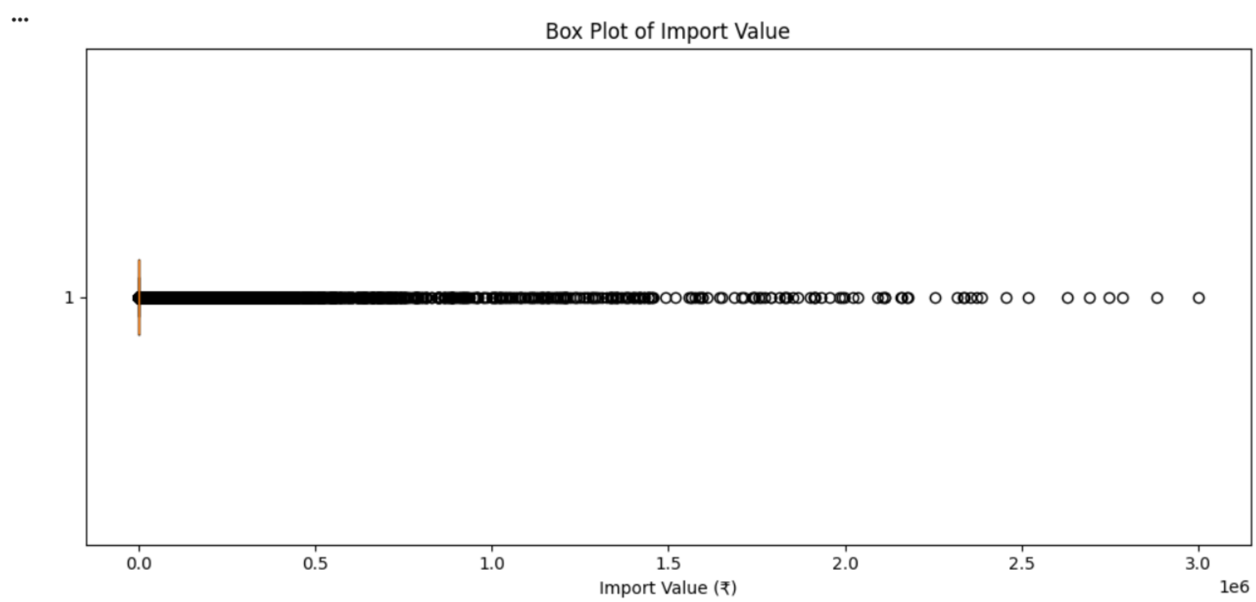
The mean being much higher than the median shows that a relatively small number of high-value observations strongly influence the average. Extreme observations should not automatically be treated as errors because they may represent genuine high-value trade records.

10. Visualizations and Findings

10.1 Import Value Box Plot

Purpose: Examines spread and extreme observations.

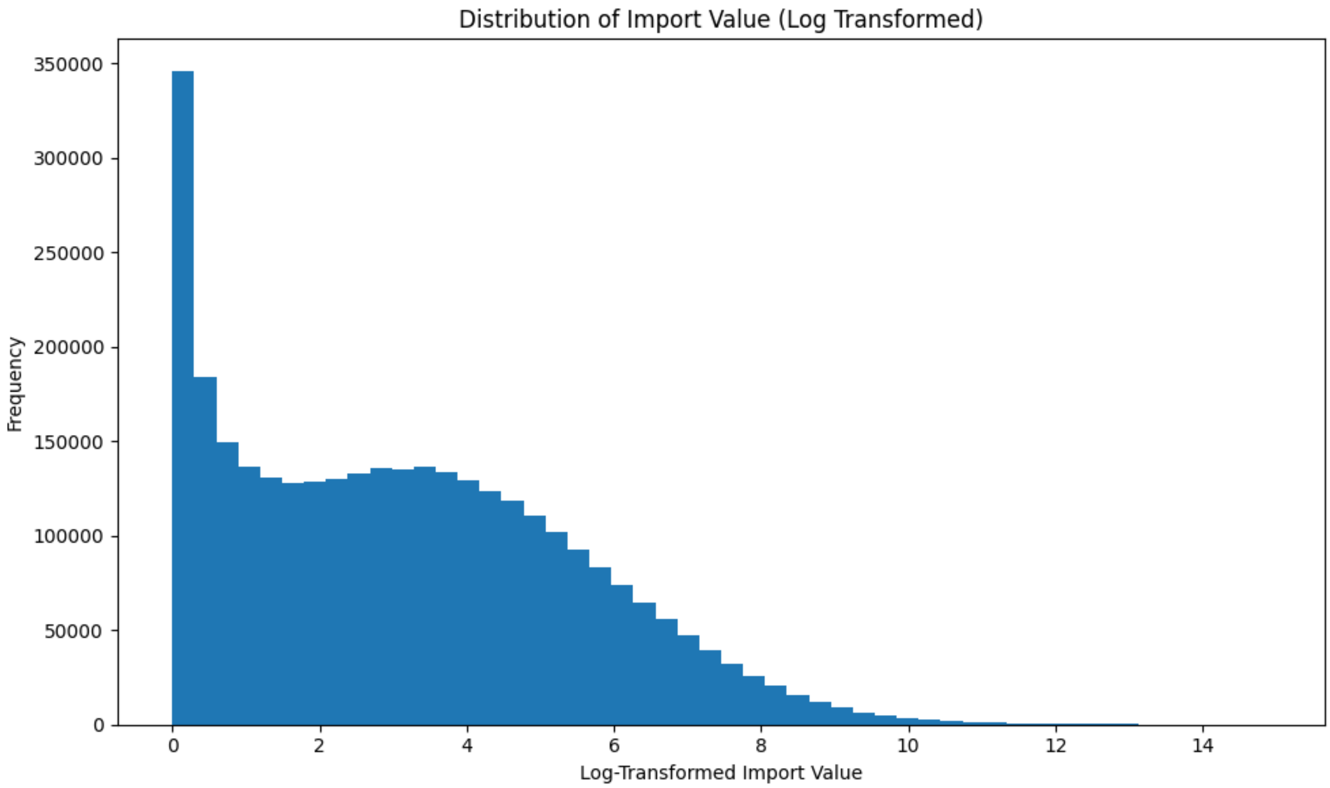
Interpretation: The plot shows substantial variation and extreme high-value observations, supporting the statistical findings.



10.2 Log-Transformed Import Value Distribution

Purpose: Improves visibility of the highly skewed monetary distribution.

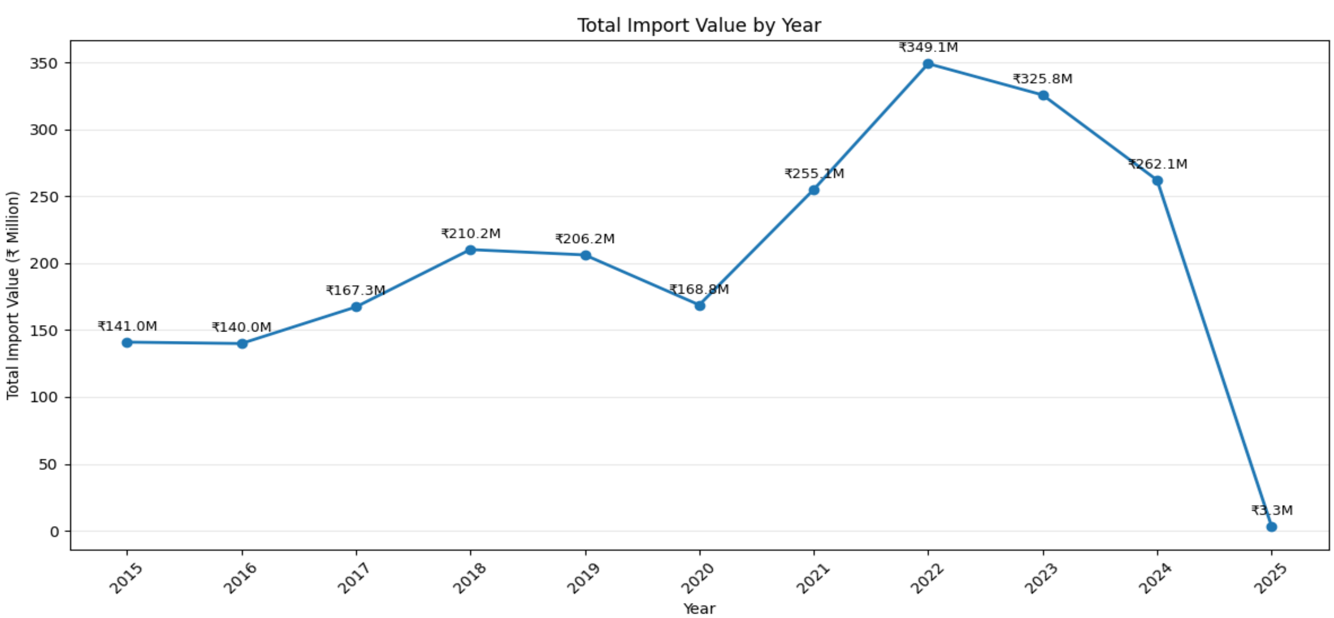
Interpretation: The log transformation makes the underlying distribution easier to inspect.



10.3 Total Import Value by Year

Purpose: Analyzes change in import value from 2015–2025.

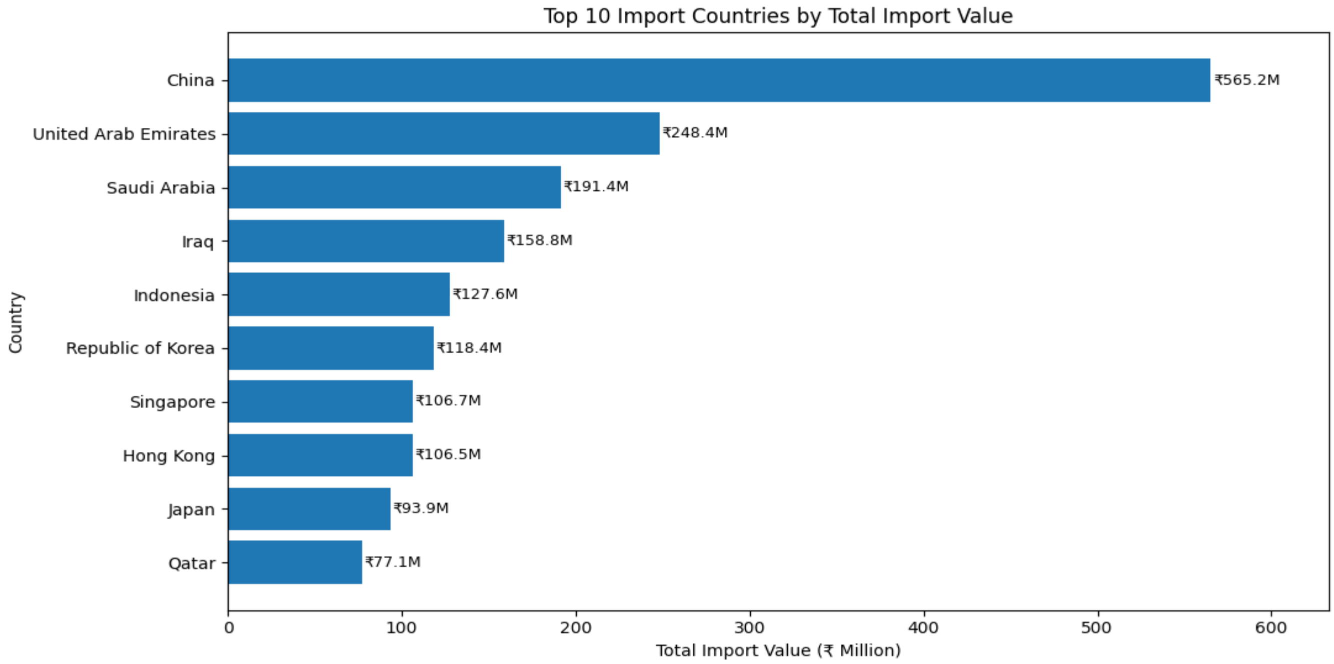
Interpretation: Import value increased from about ₹141.00M in 2015 to a peak of ₹349.15M in 2022, then declined in 2023 and 2024. 2025 is unusually low and requires validation.



10.4 Top 10 Import Countries

Purpose: Identifies countries with the largest total import values.

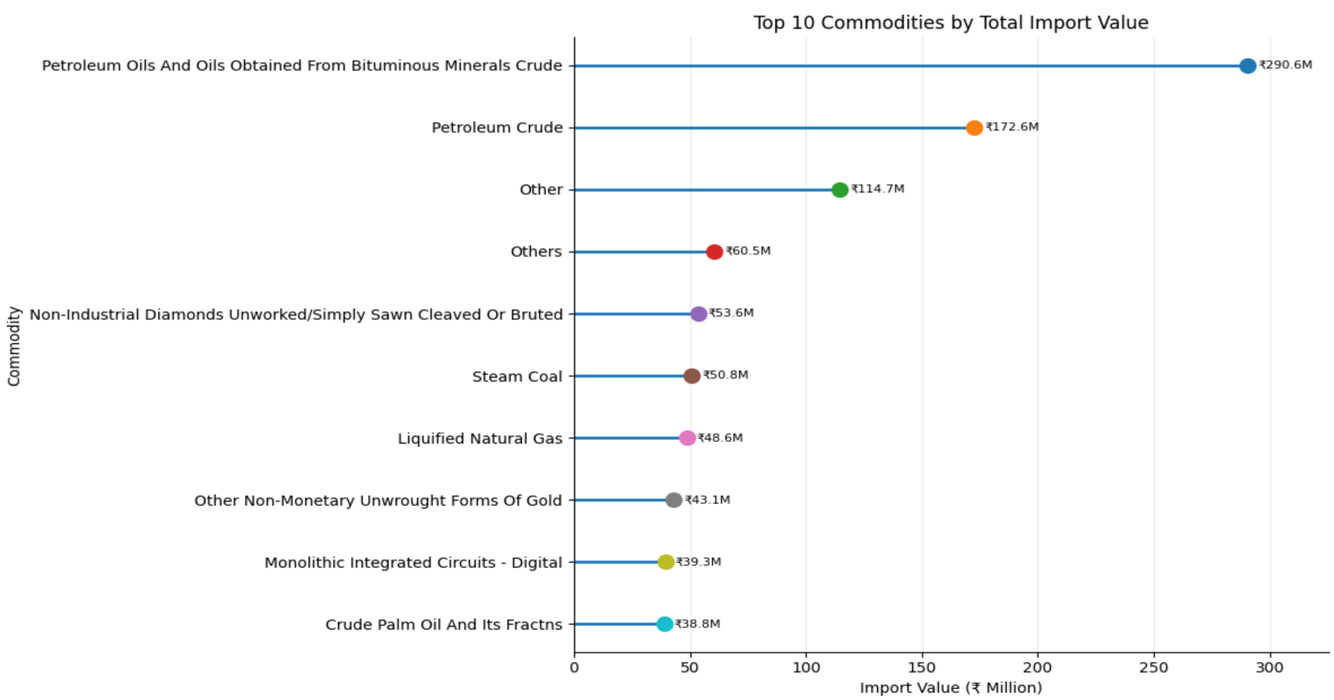
Interpretation: China is the leading major import source, followed by the UAE, Saudi Arabia and Iraq.



10.5 Top 10 Commodities

Purpose: Identifies the largest commodity contributors.

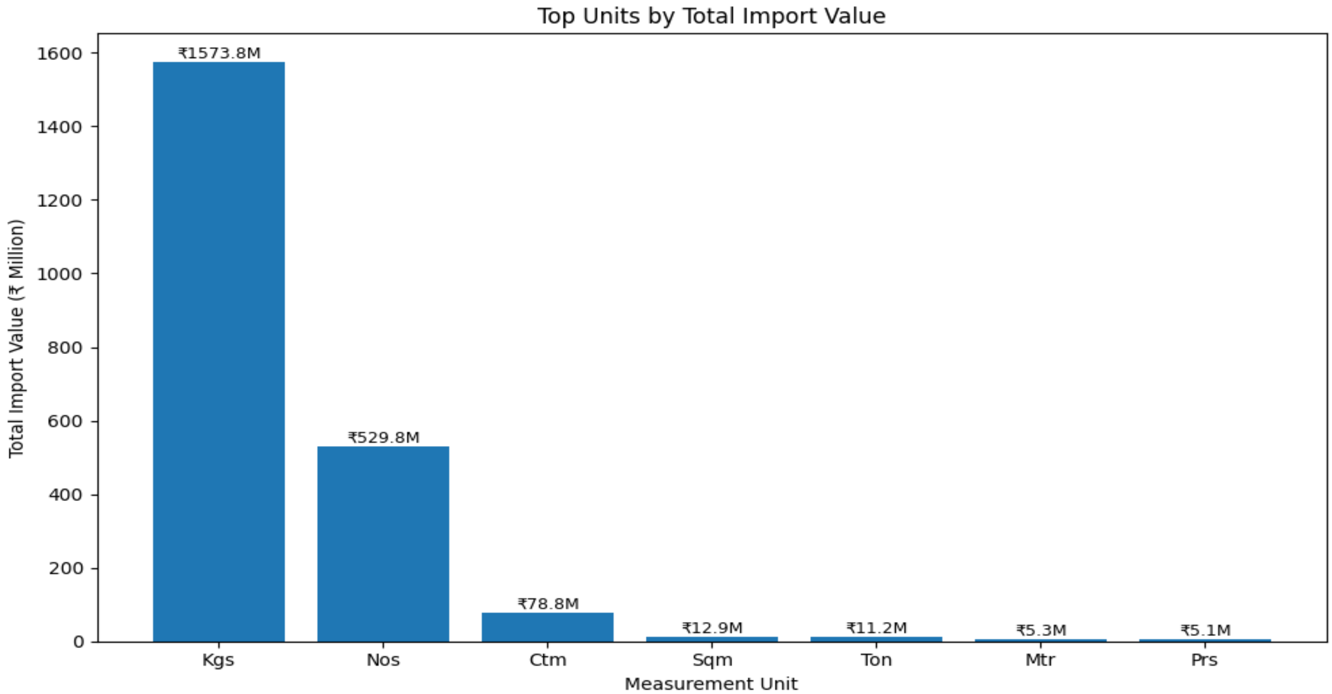
Interpretation: Petroleum oils and petroleum crude are among the largest contributors; diamonds, coal, LNG, gold and integrated circuits also appear among major contributors.



10.6 Unit-wise Import Value

Purpose: Compares monetary value across reported unit categories.

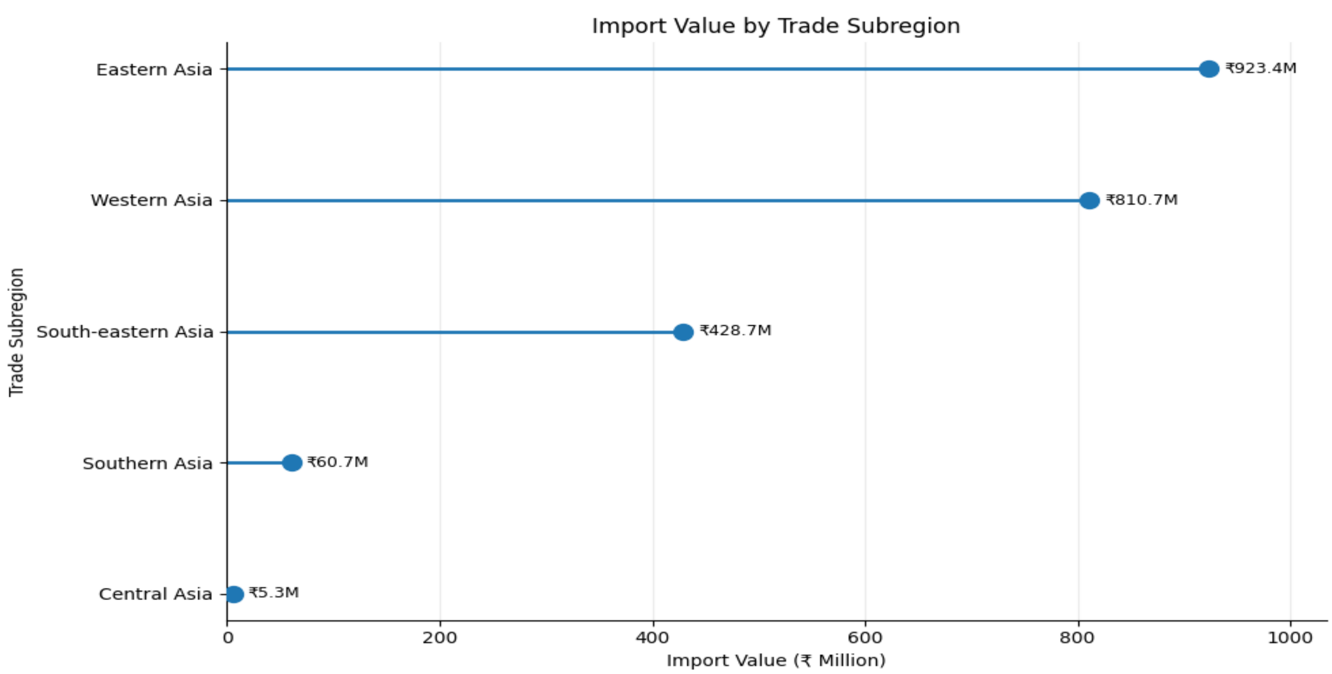
Interpretation: Value is concentrated in a few unit categories, particularly Kgs and Nos. This is a value comparison, not a physical-quantity comparison.



10.7 Import Value by Trade Subregion

Purpose: Compares import activity across Asian subregions.

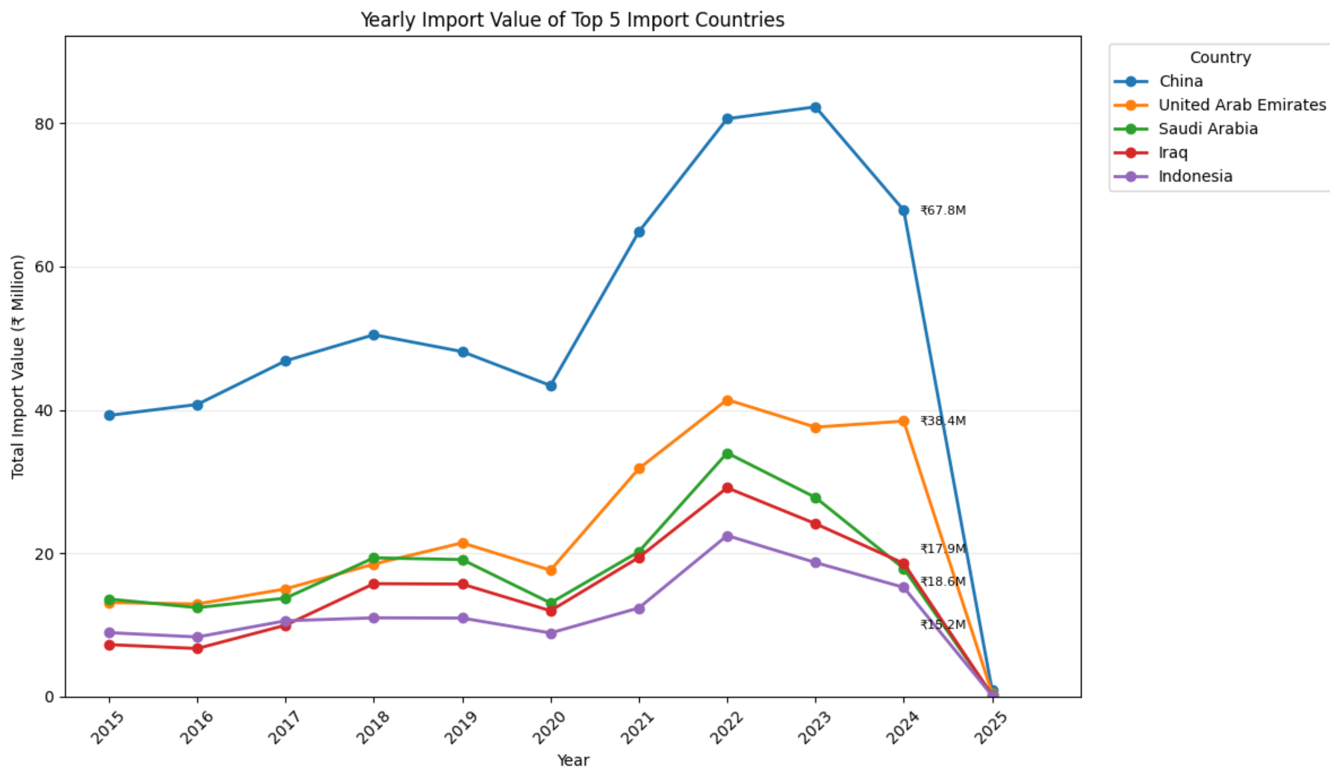
Interpretation: Import activity varies across subregions, indicating geographic concentration rather than an even distribution.



10.8 Yearly Import Value of Top 5 Countries

Purpose: Compares major country contributions over time.

Interpretation: China remains the leading contributor across most of the period, while the UAE shows strong growth. The five major countries also show unusually low values in 2025.



11. Key Insights

- Import value increased from approximately ₹141.00M in 2015 to ₹349.15M in 2022.
- China is the leading major import source.
- Petroleum-related commodities contribute substantially to total import value.
- Import values are highly uneven, with a long right tail and extreme observations.
- Import activity is concentrated across major countries and Asian subregions.
- The 2025 value is unusually low and should be validated before long-term conclusions are made.

12. Decision Support

- Monitor major trading partners such as China and the UAE.
- Track high-value commodities, particularly petroleum and energy-related products.
- Investigate major year-to-year fluctuations.
- Review extreme observations before treating them as errors.
- Validate 2025 data before strategic use.

13. Types of Analysis Performed

13.1 Descriptive Analysis

- Review extreme observations before treating them as errors. Analyzed **3,188,475 records** across **2015–2025**, including countries, commodities, units and yearly import values.
- Import value had a **mean of ₹699.02**, **median of ₹18.42**, and **standard deviation of ₹14,421.73**, showing high variation in trade values.

13.2 Diagnostic Analysis

- Compared the **mean ₹699.02 vs median ₹18.42** and found strong positive skewness (**90.41**), indicating the presence of very high-value records.
- Identified **2022 as the peak year at ₹349.15M** and observed an unusually low total of **₹3.30M in 2025**, requiring further validation.

13.3 Predictive Analysis

- **Predictive modelling was not performed** because the project focused on historical trade analysis.
- Forecasting import values by **country, commodity and year** is proposed as future work.

13.4 Decision-Support Analysis

- Identified **China as the leading import source** and petroleum-related commodities as major contributors to import value.
- The findings can support **country monitoring, commodity tracking and investigation of unusual yearly patterns**, particularly the 2025 values.

14. Limitations & Future Scope

- Quantity values use different measurement units and cannot be directly aggregated across units.
- Import values are highly skewed and contain extreme observations.
- “Other” and “Others” are source-level categories and should not be interpreted as specific commodities.
- The 2025 pattern requires validation.
- The analysis is descriptive and diagnostic rather than predictive.

15. Future Scope

- Predictive forecasting of import trends.
- Anomaly detection for unusual trade records.
- Detailed HS-code analysis.
- Country–commodity relationship analysis.
- Automated reporting and data refresh.
- Interactive Power BI dashboard development.
- Extension with newer validated data.

16. Conclusion

This project demonstrates how Python-based data analytics can transform a large import-trade dataset into meaningful insights. Through exploration, cleaning, transformation, feature engineering, statistical analysis and visualization, the project identifies major trading partners, high-value commodities, yearly trends, geographical concentration and statistical variation.

The analysis also highlights areas requiring further investigation, particularly the highly skewed monetary distribution and unusually low 2025 values. Overall, the project demonstrates an end-to-end analytical workflow from raw data to interpretable, decision-oriented findings.

17. Project Links

GitHub Repository: Link

LinkedIn: Link

Google Colab Notebook: Link